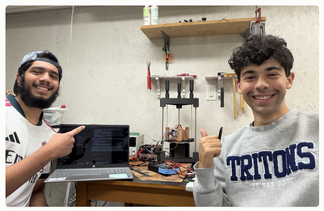
INSTRUMENT DEVELOPMENT · SCANNING PROBE MICROSCOPY

We built a scanning tunnelling microscope in a basement

Every subsystem a tunnelling microscope needs is built, working and calibrated. The amplifier resolves picoamps, the measurement chain agrees with theory to 0.13 of a standard error, and a real tip-and-sample junction responds to voltage and to height. **What is left is mechanical — and we measured it rather than guessing.**

Jacob Katz — University of California San Diego · **Nuh Shaheer** — Worcester Polytechnic Institute

Self-directed work: no faculty advisor, no grant and no laboratory. Mechanics, controller board and firmware follow Mech Panda's open-source *red-panda-stm*; the scan head and transimpedance preamplifier follow Dan Berard's home-built STM. First power-on 29 August 2026, last bench session 19 September 2026, 27 session logs, more than 160 raw data files, 92 photographs of our own hardware and every analysis script are committed to the project repository — including the results



The team, at the bench. Nuh Shaheer left, Jacob Katz right; the instrument is behind them, still assembled. Identification **SAID** by Jacob, 2026-09-19, who also permitted this frame to be published. Images/ours/2026-09-19_team_at_bench_1.jpg (IMG_8633, 09:38:09 camera-local)

Two first-year students. A basement. Our first soldering.

Neither of us writes code, and neither of us had much electronics experience — this project was the first real soldering either of us has done. It was built and run in the basement of a house people live in, with people moving around and no control over the air or the temperature: a wooden bench on a concrete floor, a few metres from the breaker panel. **Not a lab. Not a clean room.** *(Stated by Jacob Katz, 2026-09-19.)*

This is not a disclaimer. It is the result. A tunnelling microscope is normally a clean-room instrument — commercial ones sit on air tables in temperature-controlled labs — and **the blocker we found and measured is exactly the class of problem those rooms exist to remove.** Stated up front, our limitation stops being a shortfall and becomes a measurement of what this environment costs you.

0.13 σ
how far the whole measurement chain lands from what Ohm's law predicted before the run
MEASURED 2026-09-16 · panel 2

4 pA
the amplifier's own input current, against the ~1 nA it has to find — 250 times smaller
MEASURED 2026-09-15

16–59 MΩ
resistance of a real tip-and-sample junction across our current–voltage sweep
MEASURED 2026-09-17 · panel 3

131 → 132 pA
spread of one reading with the room still, then with someone stamping two metres away
MEASURED 2026-09-19 · tip clear

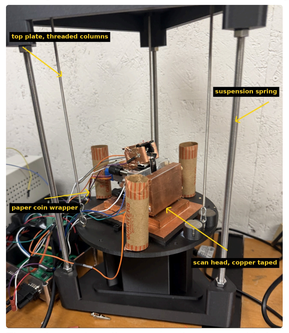
≥ 43,000
Z counts the gap moved in 6.4 seconds, motor stopped, nobody touching it. **The blocker**
MEASURED 2026-09-19 · panel 5

0 images
51 recordings, 209 reproducibility tests, no demonstrable image — our own controls are how we know
2026-09-19 · panel 6

1 What we built

There was no kit. We designed, ordered, printed, assembled, wired and characterised a complete scanning-probe measurement chain, and brought up every stage on its own before trusting the next one.

A Teensy 4.1 drives four 16-bit converters over a 1 MHz serial bus and a 26-way ribbon — X, Y and the sample bias at ±3 V, Z at ±10 V — moving a piezoelectric disc that carries the tip. Tip current goes into an OPA627 transimpedance amplifier with a 100 MΩ feedback resistor, in a hand-folded copper box as close to the tip as it physically fits, and is read back by a 16-bit LTC2326-16. Coarse approach is a 28BYJ-48 stepper on a lever. Frame, head and isolation stack are 3D-printed; the suspended platform hangs on three springs over an eddy-current damping stack. All control and analysis is Python on a laptop over USB.



The instrument, on the morning it came apart to be moved. Printed frame, a top plate on threaded columns, long fine springs down to the circular suspended platform, the copper-taped scan head on a copper-covered base plate, and paper coin



Where the work happened. A wooden bench on a concrete basement floor, a few metres from a wall-mounted electrical panel and a standby-generator transfer switch. IMG_8631, 2026-09-19 09:35:33. READ from the frame — context for the vibration problem, not a measurement of it.

2 The chain works end to end

We took the tip and the sample out and clipped a **known 100 MΩ resistor** where the junction would be, then swept the bias across it. That makes the current something Ohm's law fixes **before the run** — so the instrument is checked against a number it had no part in choosing.

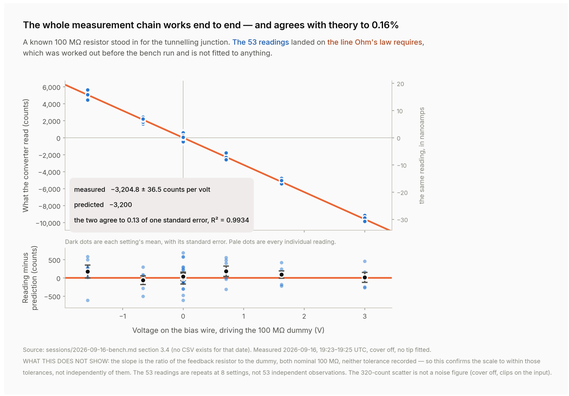


Figure 1. Measured −3,204.8 ± 36.5 counts per volt against −3,200 predicted — agreement to 0.13 of one standard error, R^2 = 0.9934, 53 readings at 8 bias settings. Twelve stages had to be right at once for those points to land on that line, from the bias converter through the wire, the tip holder and the amplifier to the firmware and the PC tools. It also settles the sign: a positive voltage on the sample reads negative. Source: sessions/2026-09-16-bench.md §3.4, measured 2026-09-16; re-fitted twice from the 53 raw readings for this poster. **What it does not show:** the slope is the ratio of two nominally 100 MΩ resistors whose tolerances were never recorded, so it confirms the scale to within those tolerances rather than independently of them; and 53 readings are repeats at 8 settings, not 53 independent observations.

3 A real junction, with the shape a barrier requires

Getting a current is easy — push the tip into the sample and there is plenty. The hard part is showing you have a **barrier**: two conductors with something in between that the electrons have to cross.

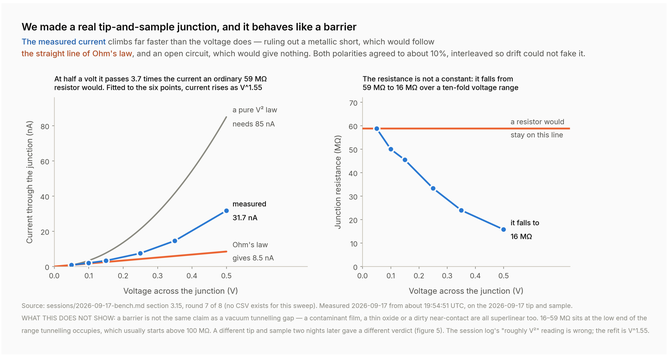


Figure 2. Current rose from 0.85 nA at 0.05 V to 31.7 nA at 0.5 V — 3.7 times faster than Ohm's law, fitting $V^{1.55}$, with the junction resistance falling from 59 to 16 MΩ. Eight rounds with the polarities interleaved, so drift could not manufacture a slope; symmetric both ways to about 10%; confirmed by a bias-flip test on two separate nights, and noise does not reverse with the voltage. That rules out a metallic short, which would be ohmic, and an open circuit, which would give nothing. Source: sessions/2026-09-17-bench.md §3.15, measured 2026-09-17. **What it does not show:** a barrier is not the same claim as a vacuum tunnelling gap — a contaminant film, a thin oxide or a dirty near-contact are superlinear too.

And the electronics are not what limits us — established with numbers rather than opinion. With no junction the chain is flat white noise at 8 to 14 counts from 2 Hz to 1.2 kHz. Leakage through the tip lead and holder is under 0.06 pA from −2 V to +2 V — more than 8 GΩ. Someone stamping on the floor two

4 What the junction is — and the ruler that settles it

What separates a tunnelling gap from a contact is **how fast the current changes as the tip moves**. We ran it: 110 cycles in and out, at four bias voltages.

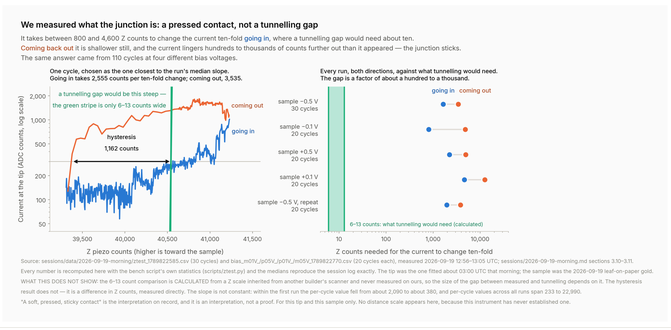


Figure 3. On the 2026-09-19 junction the current changed ten-fold per 1,650 to 1,970 Z counts going in, where a tunnelling gap on the inherited Z scale would need about 6 to 13 — and every run showed 705 to 1,868 counts of in-and-out hysteresis, a mechanical signature that depends on no distance scale and that did not shrink when we cut the bias fivefold. That junction was a pressed contact: measured, and not in doubt. Source: sessions/data/2026-09-19-morning/, measured 2026-09-19 12:56–13:05 UTC. **What it does not show:** the 6–13 figure is calculated from a Z scale borrowed from another builder's scanner and never measured on ours. The hysteresis does not depend on it.

Our best junction's status is still undecided — on one unmeasured distance. The junction in panel 3 is **a different tip and a different sample**, two nights earlier, and its current changed ten-fold about **seven times faster**: roughly 250 Z counts per decade. Whether that was tunnelling turns on **d**, how far the tip stands from the line through the two ball supports, because that sets the lever ratio between the motor and the tip. **Below about 0.13 mm it was tunnelling; at 1 mm it was resting on something. d has never been measured** — Jacob's own estimate is "between 1 mm and 0 mm", so the threshold sits inside the range

5 We found what stops us, and measured it

To take an image, the gap has to hold still while the tip is swept. We stopped the motor, took our hands off the instrument, and watched where the surface was.

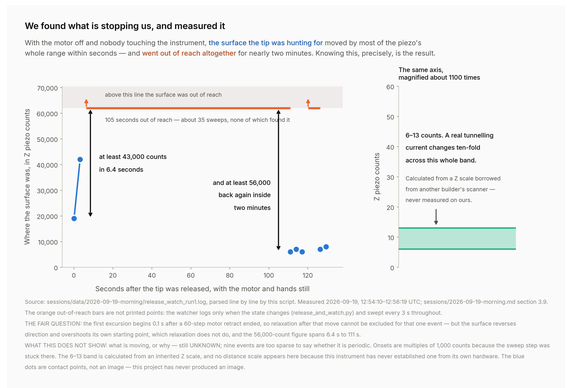


Figure 4. The surface moved by more than 43,000 Z counts within 6.4 seconds and by more than 56,000 within two minutes — most of the piezo's whole range — and went out of reach altogether for 105 seconds. Source: sessions/data/2026-09-19-morning/release_watch_run1.log, measured 2026-09-19 12:54–12:56 UTC. **The fair question:** the fast excursion begins 0.1 s after a 60-step motor retract ended, so relaxation after that move cannot be excluded for that one event — but the motion reverses and overshoots its own starting point, which relaxation does not do, and the two-minute figure is measured long after any settling transient.

Why that ends the imaging question, in one comparison. Over the five seconds one image takes, the gap drifts **396 to 1,722 counts**. The "structure" in every candidate image we have is **29 to 458 counts. The**

6 How we tested ourselves — and the open question

Every scan was immediately re-run with the sideways sweep switched off. An **X-held control** has identical timing and identical loop behaviour, but the tip never travels across the surface — **so it cannot contain surface structure at all, by construction.** Anything that appears in both is the instrument talking to itself. Like for like over full images, our scans repeat at **+0.04** and their controls at **+0.07**: indistinguishable, and neither differs from zero.

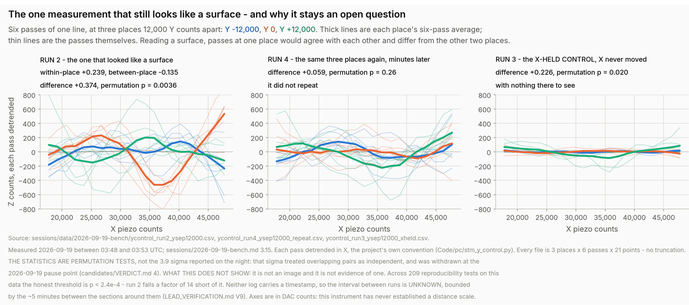


Figure 5. One measurement resists that. Holding the tip at three places 12,000 Y counts apart, passes at the same place agreed +0.239 while passes at different places disagreed −0.135 — a difference of +0.374, permutation p = 0.0036. The repeat, minutes later, gave +0.056 (p = 0.26), and an X-held control, which cannot contain a surface, scored p = 0.020 on the same statistic — this instrument's own false-positive rate, and knowing it is worth more than any single positive would have been. Across 209 reproducibility tests the honest threshold is $p < 2.4 \times 10^{-4}$, fixed before the candidates were judged. Source: sessions/data/2026-09-19-bench/ycontrol_run2, _run3, _run4. **This is not an image, and we do not present it as one.**

What keeps it open, and the experiment that closes it. The instrument's own logs show the gap was **steady** across these runs — the onset moved 2,200 counts, about 3% of the Z range — so "the surface moved out from under it" is not the explanation, and genuine non-reproduction is the better-supported reading. **But nothing in**

What we do not claim

- **No image.** Fifty-one recordings, no demonstrable image of the sample — and we know that from our own controls, not from nothing turning up.
- **No demonstrated tunnelling.** We have a barrier junction; the 2026-09-19 junction was a pressed contact, measured. Our best junction's status is **undecided**, and panel 4 names the measurement that settles it.
- **No atomic resolution, and no scale bar anywhere on this poster** — every axis is in converter counts, because this instrument has never established a distance scale of its own. **No number is taken off a photograph:** every caption is marked **SAID** or **READ**.

Where it goes next

- **Measure d with a straightedge** — no power, a few minutes, and it settles retroactively whether our best junction was tunnelling.
- **Prove the gap holds still before scanning anything:** a box over the instrument, a 15-minute stillness recording, and a gate of about 200 counts over sixty seconds.
- **A stiffer sample than gold leaf on paper, and a sharper tip.**
- **Three software fixes that cost nothing** and remove the artefact behind our most convincing false positive: drop the first three pixels of each pass, sweep X back and forth between passes, and refuse to compare two images of different length.

We owe these people — and we checked our own work

Mech Panda (*red-panda-stm*: mechanics, controller, firmware) and **Dan Berard** (scan head, transimpedance preamplifier) published their instruments for free. Without them there would have been nothing to bring up.

At this pause point we re-derived our own published numbers from the raw files and found six of them wrong — including two statistics we had used to dismiss our own results, and a reproducible feature that turned out to be the feedback loop recovering from a jump rather than the sample. **Every conclusion survived; two of the arguments for them did not.** All of it is in the repository, beside the data.